

Q1: Considering the following elemental stress distribution, obtain the required items.

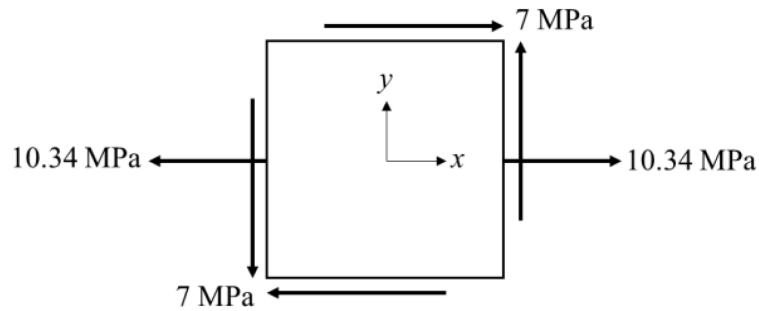
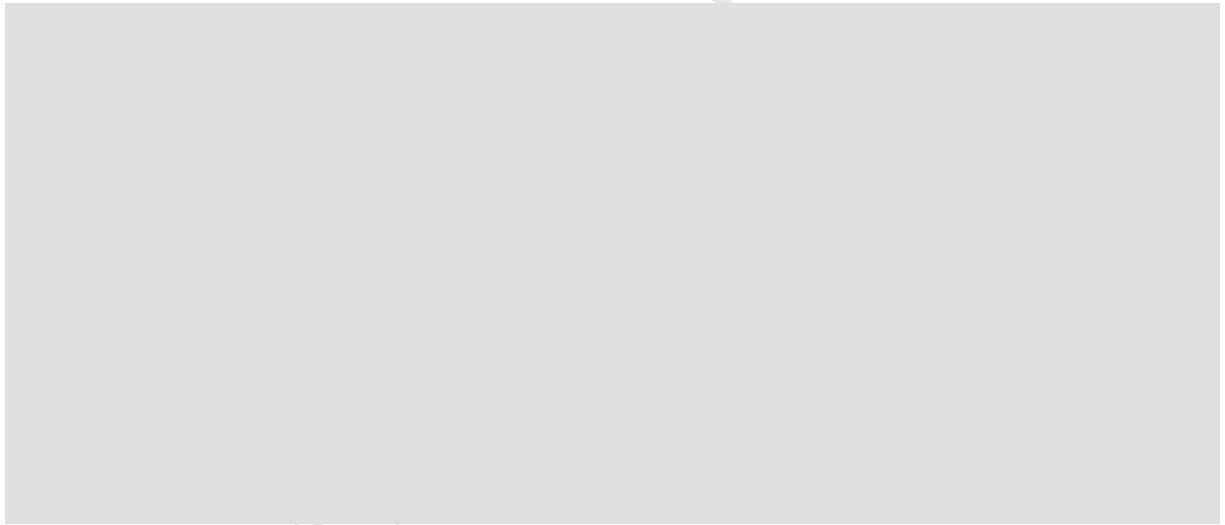


Figure 1: Q1's element

$$\begin{cases} \sigma_x = 10.34 \text{ MPa}, \sigma_y = 0 \text{ MPa}, \tau_{xy} = 7 \text{ MPa} \\ \nu = 0.3, E = 70 \text{ GPa} \end{cases}$$

a. Calculate the principal stresses and their plane by equations.



$$\sigma_1 = 13.87 \text{ MPa}, \sigma_2 = -3.53 \text{ MPa}, \tau_{max,min} = \pm 8.7 \text{ MPa}$$

$$\theta_{\sigma_1} = 26.78^\circ, \theta_{\sigma_2} = -63.22^\circ, \theta_{\tau_{max}} = 71.78^\circ, \theta_{\tau_{min}} = -18.22^\circ$$

b. Calculate the principal stresses and their plane by Mohr's circle.

$$\sigma_1 = 13.87 \text{ MPa}, \sigma_2 = -3.53 \text{ MPa}, \tau_{max,min} = \pm 8.7 \text{ MPa}$$

$$\theta_{\sigma_1} = 26.78^\circ, \theta_{\sigma_2} = -63.22^\circ, \theta_{\tau_{max}} = 71.78^\circ, \theta_{\tau_{min}} = -18.22^\circ$$

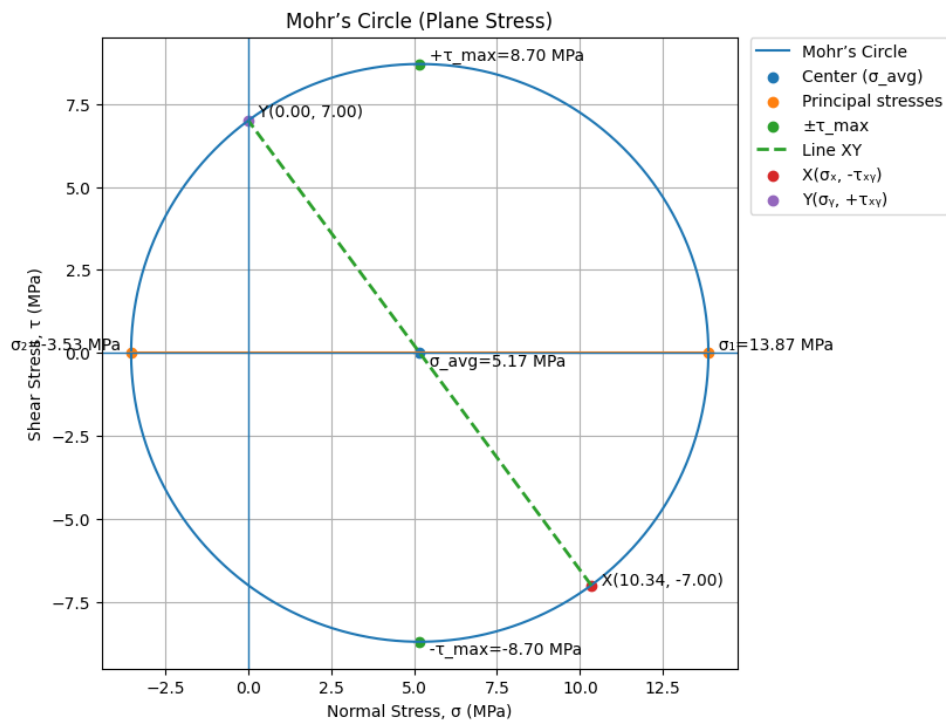


Figure 2: Mohr's circle

c. Calculating the x and y components of the normal and shear strains.

$$\varepsilon_x = 147.71 \mu, \varepsilon_y = -44.3 \mu, \gamma_{xy} = 260 \mu$$

d. Calculate the principal strains

$$\varepsilon_1 = 213.31 \mu, \varepsilon_2 = -109.91 \mu, \gamma_{max} = 323.23 \mu$$

Q2: An aluminum plate with an elastic modulus of 74 GPa and a Poisson's ratio of 0.33 is subjected to vertical loading. Before loading, a line with a slope of 1:2 is considered on the plate. Calculate the slope of the line after applying a stress of 125 MPa.

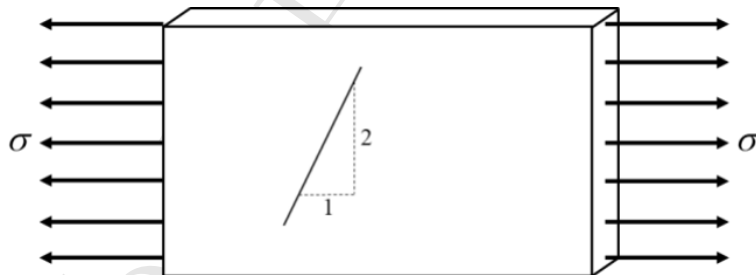


Figure 3: element's loading and slope

$$\begin{cases} \sigma_x = 125 \text{ MPa}, \sigma_y = 0 \text{ MPa}, \tau_{xy} = 0 \text{ MPa} \\ \nu = 0.33, E = 74 \text{ GPa}, \text{ slope} = 2 \text{ (1:2)} \end{cases}$$

new slope: 1.999

Q3: A circular-cross-section bar is subjected to a bending moment of 21 kN.m and a torsional moment T . At a point on this member, the measured strains correspond to relations (1–3), and their orientation is shown in Figure 3. If the elastic modulus and Poisson's ratio are equal to 200 GPa and 0.3, respectively, determine the diameter of the bar (d) and the torsional moment T .

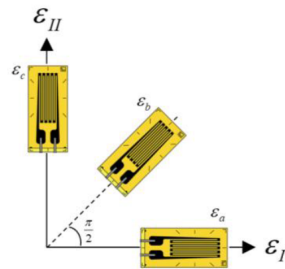


Figure 4: strain gauge

$$\begin{cases} M = 21 \text{ KN.m.}, E = 200 \text{ GPa}, \nu = 0.33 \text{ MPa} \\ \varepsilon_a = 2.17 \times 10^{-3}, \varepsilon_b = 0.22 \times 10^{-3}, \varepsilon_c = -1.73 \times 10^{-3} \end{cases}$$

$$d = 10.86 \text{ cm}, T = 35.89 \text{ KN.m}$$

Q4: The bar subjected to the applied loading shown in Figure 4 is considered. Determine the principal stresses at point A.

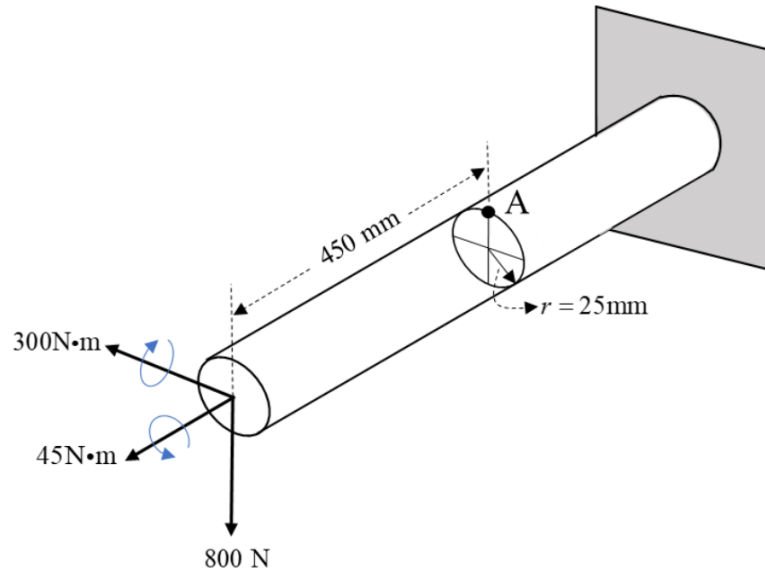


Figure 5: Loaded bar

$$\begin{cases} T_x = 45 \text{ N.m.}, M_y = -300 \text{ N.m.}, P_z = -800 \text{ N} \\ r = 25 \text{ mm}, d = 450 \text{ mm} \end{cases}$$

$$\sigma_1 = 6.8 \text{ MPa}, \sigma_2 = -1.9 \text{ MPa}, \tau_{max,min} = \pm 4.35 \text{ MPa}$$

Q5: Using the Airy stress function for the cantilever beam under a triangular load, determine the constants C_i ($i = 1, \dots, 6$) such that the biharmonic equation and all boundary conditions are satisfied. The Airy stress function is assumed in the form:

$$\phi(x, y) = C_1 xy + \frac{C_2}{6} x^3 + \frac{C_3}{6} x^3 y + \frac{C_4}{6} xy^3 + \frac{C_5}{9} x^3 y^3 + \frac{C_6}{20} xy^5.$$

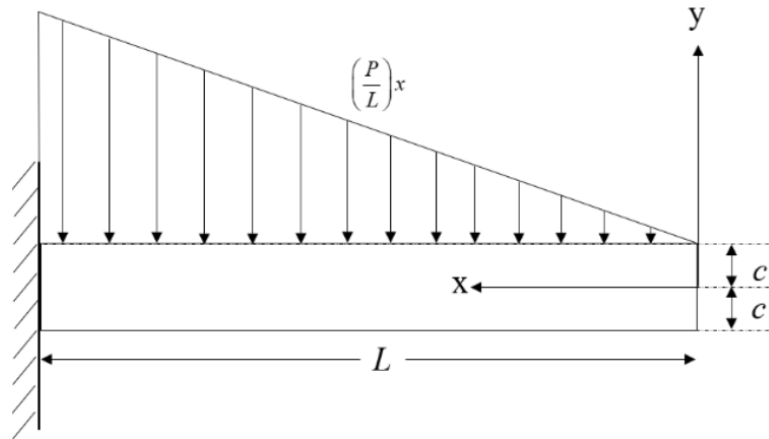
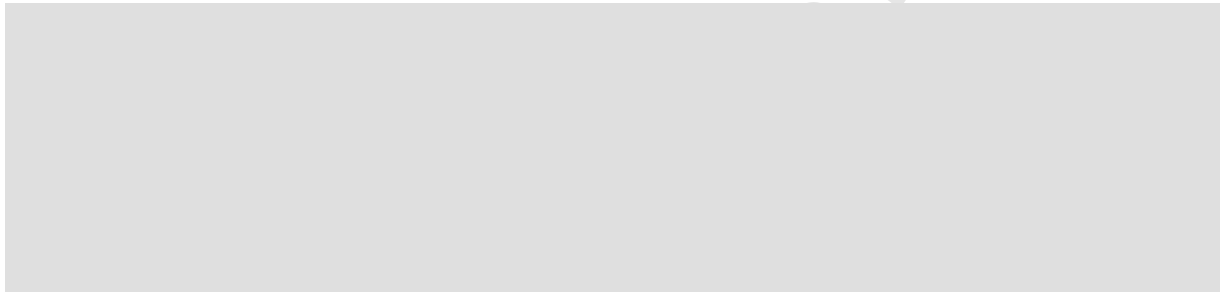
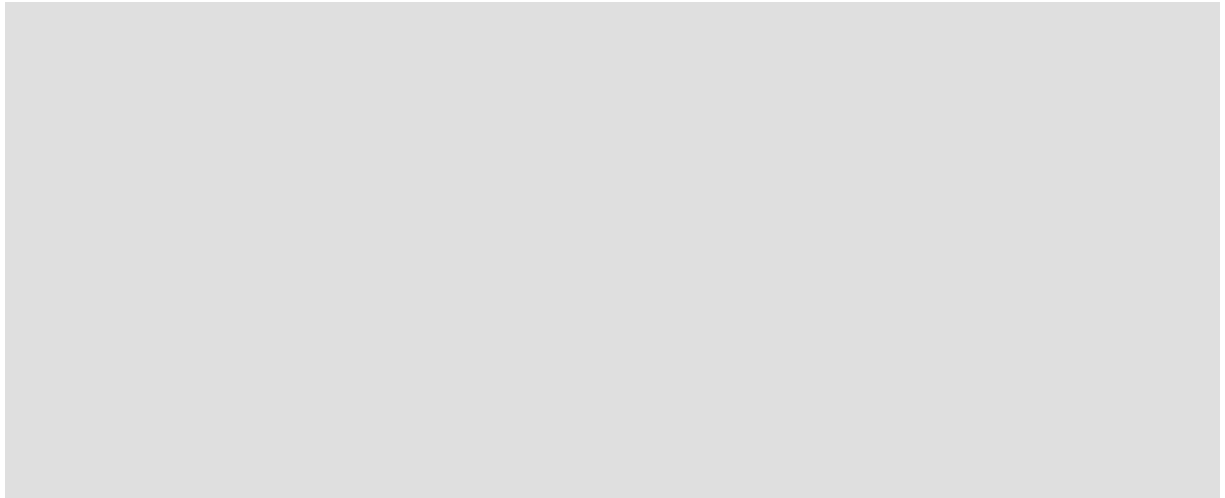


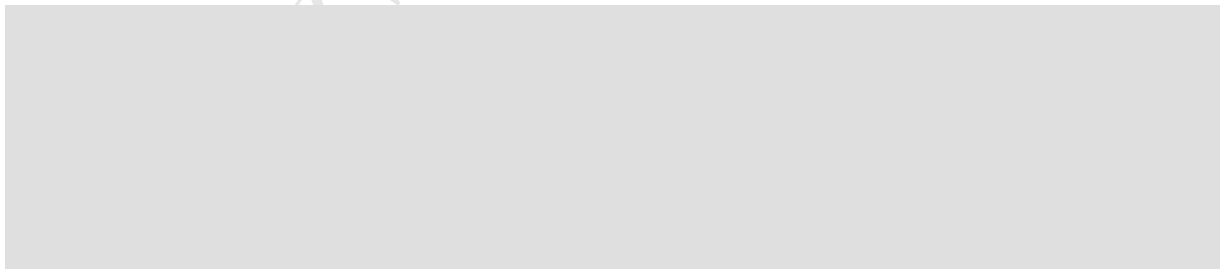
Figure 6: Loaded beam

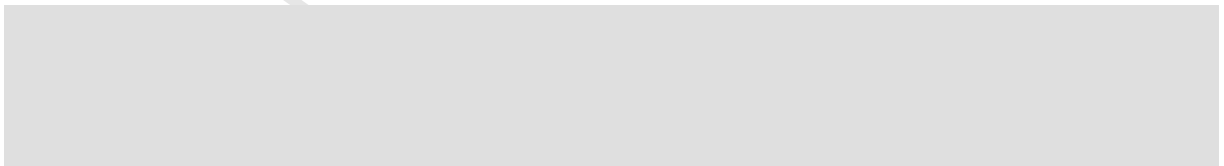
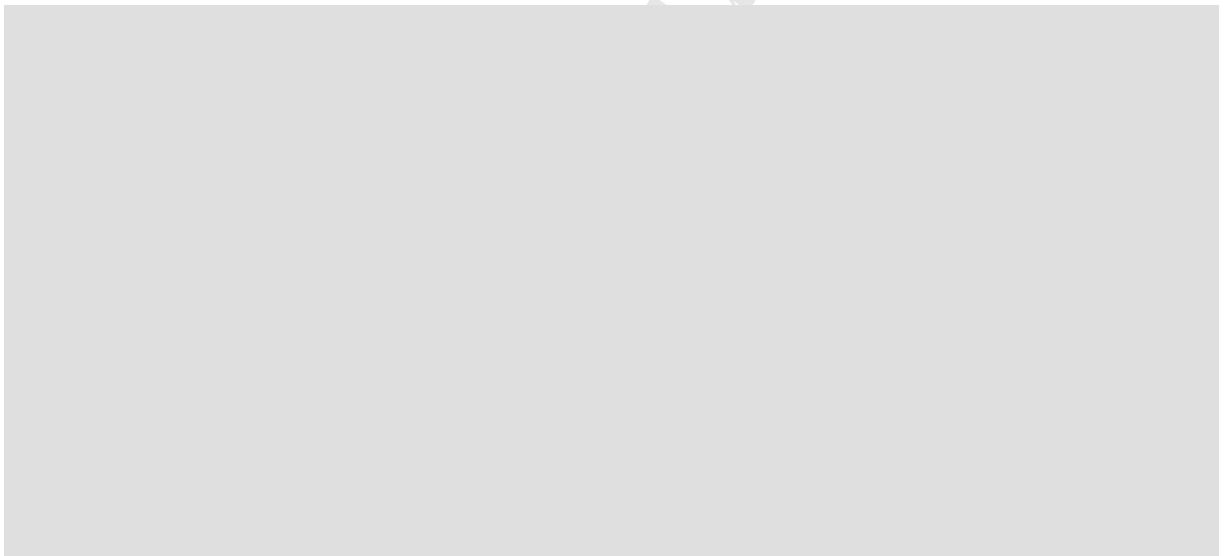
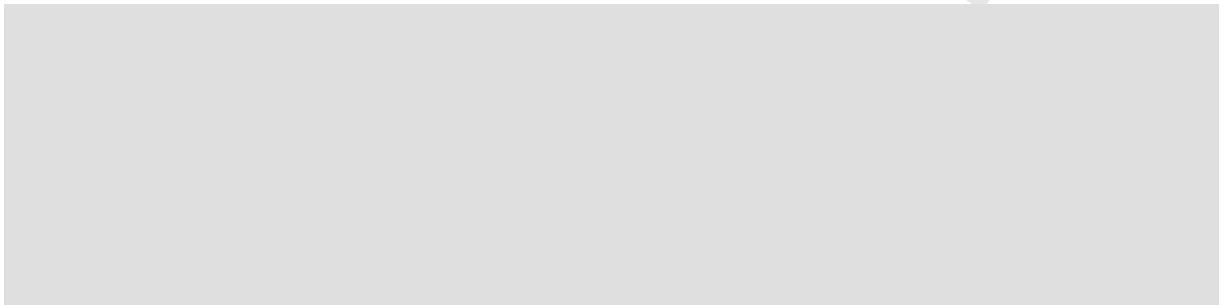
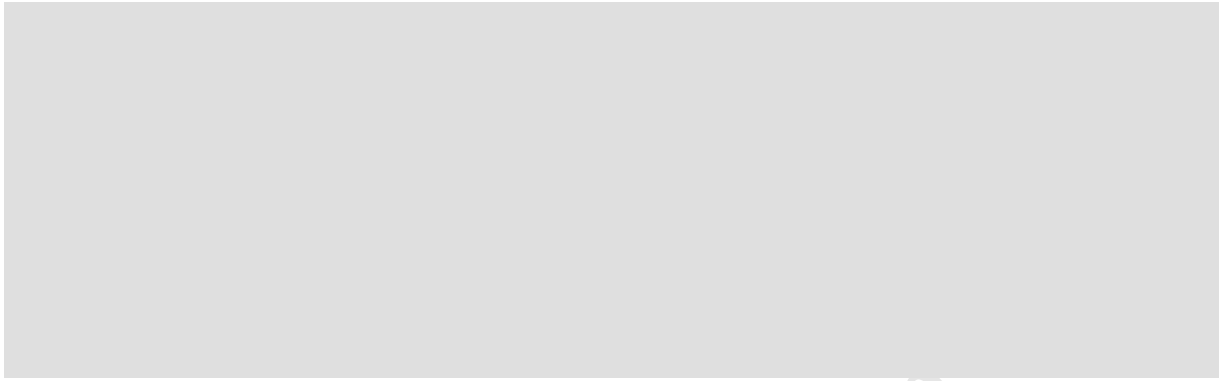


$$C_1 = -\frac{cp}{40L}, \quad C_2 = -\frac{p}{2L}, \quad C_3 = -\frac{3p}{4Lc}, \quad C_4 = \frac{3p}{10Lc}, \quad C_5 = \frac{3p}{8Lc^3}, \quad C_6 = -\frac{p}{2Lc^3}$$

Q6: Using the stress field obtained in Q3, determine the in-plane displacement components $u(x, y)$ and $v(x, y)$ for the cantilever beam, assuming plane stress and the following boundary conditions at the clamped end:

$$u(L, 0) = 0, \quad v(L, 0) = 0, \quad \left. \frac{\partial u}{\partial y} \right|_{(L, 0)} = 0$$



 $+ C_1$

Substituting these constants back into the expressions for $u(x, y)$ and $v(x, y)$ and simplifying, we obtain the final displacement field:

$$u(x, y) = \frac{p}{80ELc^3} \left[5x^4y - (10\nu + 20)x^2y^3 + (30c^2\nu + 12c^2)x^2y + 20c^3\nu x^2 \right. \\ \left. + (2\nu + 3)y^5 + (-4c^2\nu + 2c^2)y^3 + 20c^3y^2 \right. \\ \left. - (5L^4 + 30L^2c^2\nu + 12L^2c^2)y - 20L^2c^3\nu \right]$$

$$v(x, y) = \frac{p}{80ELc^3} \left[-x^5 - 10\nu x^3y^2 + (10c^2\nu + 16c^2)x^3 + (10\nu + 5)xy^4 \right. \\ \left. + (-12c^2\nu - 30c^2)xy^2 - 40c^3xy \right. \\ \left. + (5L^4 + 30L^2c^2\nu + 12L^2c^2 + 4c^4\nu + 4c^4)x \right. \\ \left. - 4L^5 - 40L^3c^2\nu - 28L^3c^2 - 4Lc^4\nu - 4Lc^4 \right]$$

Q7: Find coefficients of the Airy stress function for the ring's shown in the figure.

$$\phi(r, \theta) = a_1 \ln r + a_2 r^2 + (a_{21} r^2 + a_{22} r^4 + a_{23} r^{-2} + a_{24}) \cos 2\theta$$

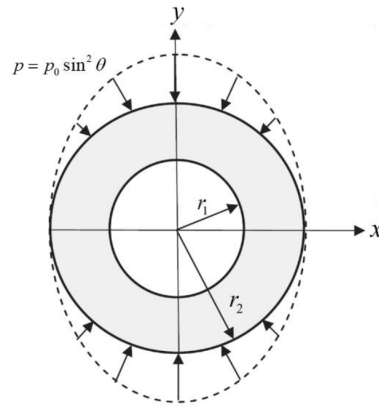


Figure 7: Ring condition

$$\begin{aligned}
 a_1 &= \frac{p_0 r_1^2 r_2^2}{2\Delta}, \quad a_2 = -\frac{p_0 r_2^2}{4\Delta} \\
 a_{21} &= -\frac{p_0 r_2^2 (2r_1^4 + r_1^2 r_2^2 + r_2^4)}{4\Delta^3}, \quad a_{22} = \frac{p_0 r_2^2 (3r_1^2 + r_2^2)}{12\Delta^3} \\
 a_{23} &= -\frac{p_0 r_1^4 r_2^4 (r_1^2 + 3r_2^2)}{12\Delta^3}, \quad a_{24} = \frac{p_0 r_1^2 r_2^2 (r_1^4 + r_1^2 r_2^2 + 2r_2^4)}{4\Delta^3}
 \end{aligned}$$

Q8: Consider a bar with the cross-section. Determine the stress distribution, the rate of twist, and the warping of the cross-section.

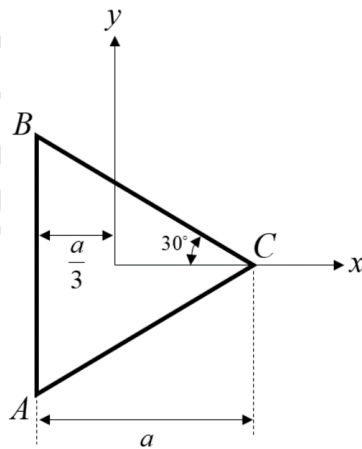
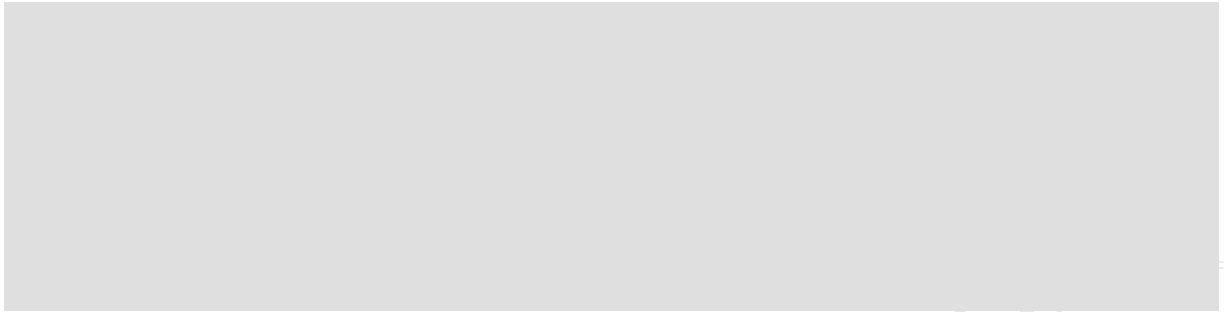
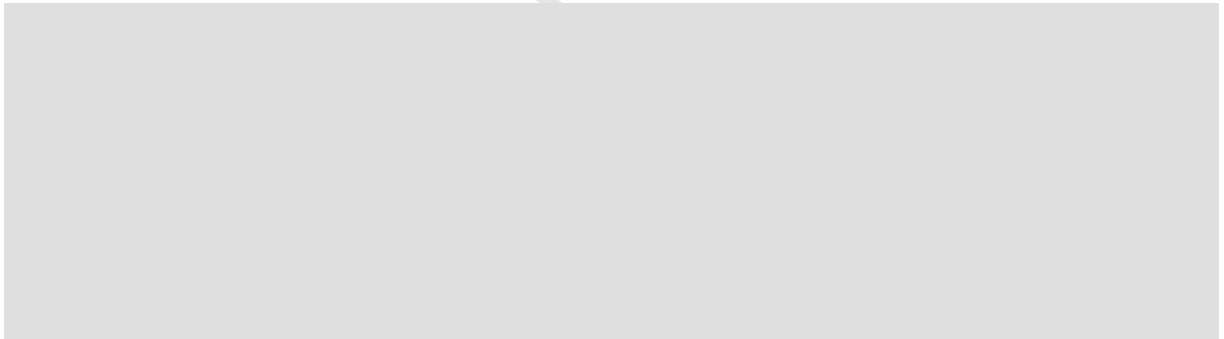
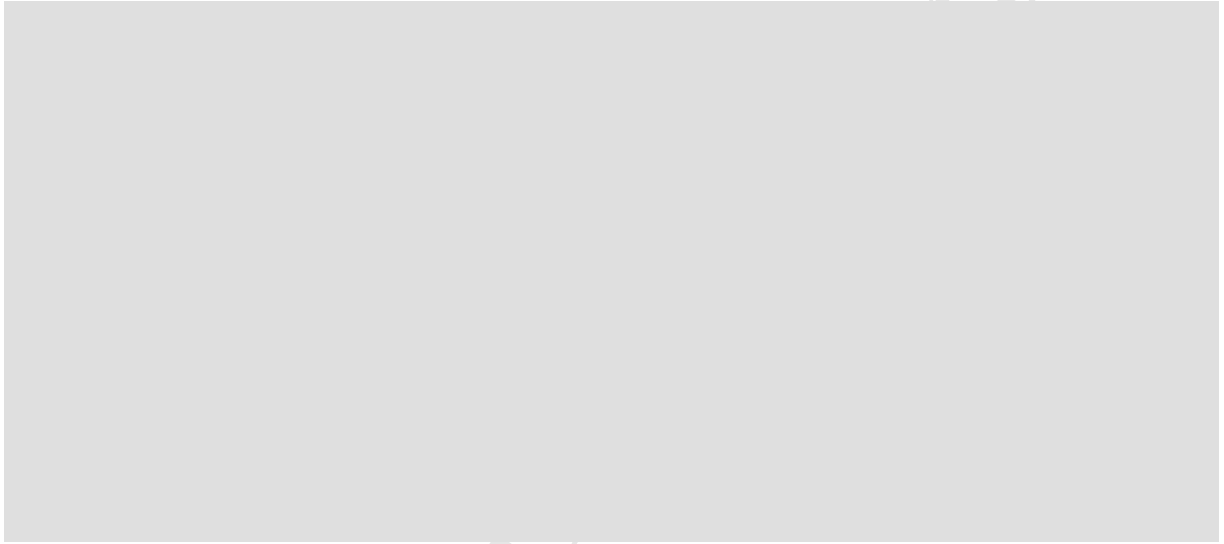


Figure 8: Bar cross-section

$$\phi = -G \frac{d\theta}{dz} \left[\frac{1}{2}(x^2 + y^2) - \frac{1}{2a}(x^3 - 3xy^2) - \frac{2}{27}a^2 \right]$$



= 0





$$\tau_{xy} = G \frac{d\theta}{dz} \left(x - \frac{3x^2}{2a} + \frac{3y^2}{2a} \right)$$

$$\tau_{zx} = -G \frac{d\theta}{dz} \left(y + \frac{3xy}{a} \right)$$

$$\tau_{xy_{max}} = -\frac{Ga}{2} \frac{d\theta}{dz}$$

$$\frac{d\theta}{dz} = \frac{15\sqrt{3}T}{Ga^4}$$

$$w = \frac{1}{2a} \frac{d\theta}{dz} (y^3 - 3x^2y)$$